

Food-washing monkeys recognize the law of diminishing returns

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Abstract

Few animal species have the cognitive faculties or prehensile abilities needed to eliminate costly tooth-damaging grit from food surfaces. Some populations of monkeys wash sand from foods when standing water is readily accessible, but this propensity varies within groups for reasons unknown. Spontaneous food-washing emerged recently in a group of long-tailed macaques (*Macaca fascicularis*) inhabiting Koram Island, Thailand, and it motivated us to explore the factors that drive individual variability. We measured the mineral and physical properties of contaminant sands and conducted a field experiment, eliciting 1,282 food-handling bouts by 42 monkeys. Our results verify two long-standing presumptions, that monkeys have a strong aversion to sand and that removing it is intentional. Reinforcing this result, we found that monkeys clean foods beyond the point of diminishing returns, a suboptimal behavior that varies with rank. Dominant monkeys abstain from washing, balancing the long-term benefits of mitigating tooth wear against immediate energetic requirements, an essential predictor of reproductive fitness.

eLife assessment

This **valuable** study tests the functional role of food-washing behavior in removing tooth-damaging sand and grit in long-tailed macaques and whether dominance rank predicts the level of investment in the behavior. The evidence that food-washing is deliberate is **compelling**, but the evidence for variable and adaptive investment depending on rank is **incomplete** given confounding between sex and rank and limited sample size. A more careful and perhaps restrained interpretation of the findings, as well as a connection to the existing literature on optimal foraging theory, would increase the value of the study to its intended audience, i.e. researchers interested in foraging behavior, cognition, and primate evolution.

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Introduction

Koshima Island, Japan is a storied fieldsite in the annals of primatology. Observations of wild macaques (*Macaca fuscata*) began there in 1948, but four years of sustained effort failed to habituate the monkeys. In August 1952, Itani and Tokuda (1954) resorted to scattering wheat grains and sweet potatoes on oft-used paths, gradually shifting the provisions to a sandy beach at Otomari Bay, a strategy that afforded a clear view of the entire group of 22 animals. Individual identifications followed quickly, laying the foundation for decades of influential research. In September 1953, a young female named Imo gathered a sweet potato from the beach and washed it in a freshwater stream, a behavioral innovation that spread horizontally to peers and then vertically to older kin (Kawai, 1965). By 1958, sweet potato washing had become a group-wide trait with a key modification: the monkeys began using sea water instead of standing freshwater, a preference that continues today seven generations later (Hirata et al., 2001).

These events have since passed into canon as an example of socially transmitted behavior, or culture, among nonhuman primates (McGrew, 1998; Matsuzawa and McGrew, 2008; Matsuzawa, 2015). But our preoccupation with culture has overshadowed fundamental questions of motivation (Sarabian and MacIntosh, 2015; Fiore et al., 2020). Two tacit assumptions—that sand produces an objectionable sensation on teeth, and that it is prudent to minimize tooth damage—are sufficiently intuitive that formal tests are wanting (Fannin et al., 2021). In consequence, the mineral and physical properties of contaminant sands are unknown, let alone the efficiency of different cleaning behaviors (Schofield et al., 2018). Another enigma concerns the preference of some monkeys to brush food on themselves (Kawai et al., 1992), a rapid but seemingly inferior means of sand removal. Food-brushing individuals have been characterised as inept (Kawai, 1965) or subordinate (Watanabe, 1994) in part because the quartz in sand can cause severe tooth damage (Lucas et al., 2013; Towle et al., 2022). Yet, carrying food to the ocean is expected to incur energetic costs as well as opportunity costs, factors that impelled us to explore the trade-offs of mitigating sand-mediated tooth wear.

Koram Island, Thailand

The long-tailed macaques (*M. fascicularis*) of Koram Island, Thailand use stone tools to harvest shellfish, a phenomenon that came to light during biodiversity damage assessments following the Sumatra-Andaman earthquake and tsunami of December 26, 2004 (Malaivijitnond et al., 2007; Gumert and Malaivijitnond, 2012; Tan et al., 2015). The monkeys are now a magnet for tourists, who provide them with market-sourced fruits (cucumbers, melon, pineapple), jettisoning them onto the beach. These events produce food surfaces with considerable concentrations of sand ($\bar{x} = 3.7 \pm 1.3 \text{ mg mm}^{-2}$) and elicit food-washing and food-brushing behaviors among the monkeys. To understand the factors driving these reactions, we examined the mineral properties of foodadhering sands ($n = 758$ particles), finding that 78% of our sample was composed of crystalline quartz (Figure 1A).

Harder than enamel, quartz can exact a heavy toll on teeth, but the probability and degree of enamel loss is governed partly by the size and shape of individual particles, factors that determine the ‘attack angle’ during particle-enamel contact (Lucas et al., 2013). We calculated the circularity of particles as a convenient proxy for sphericity (Grace and Ebneyamini, 2021), finding that it decreased as a function of particle size (Figure 1B). This result suggests that larger particles are more angular, posing a greater risk to enamel. However, we also calculated a median Feret diameter of 25.8 μm (range: 8.7 to 644 μm), meaning that nearly half the sample existed below the human threshold (25 μm) of oral detection (Imai et al., 1995). To put 25 μm into perspective, it is one-twelfth the diameter of the period ending this sentence.

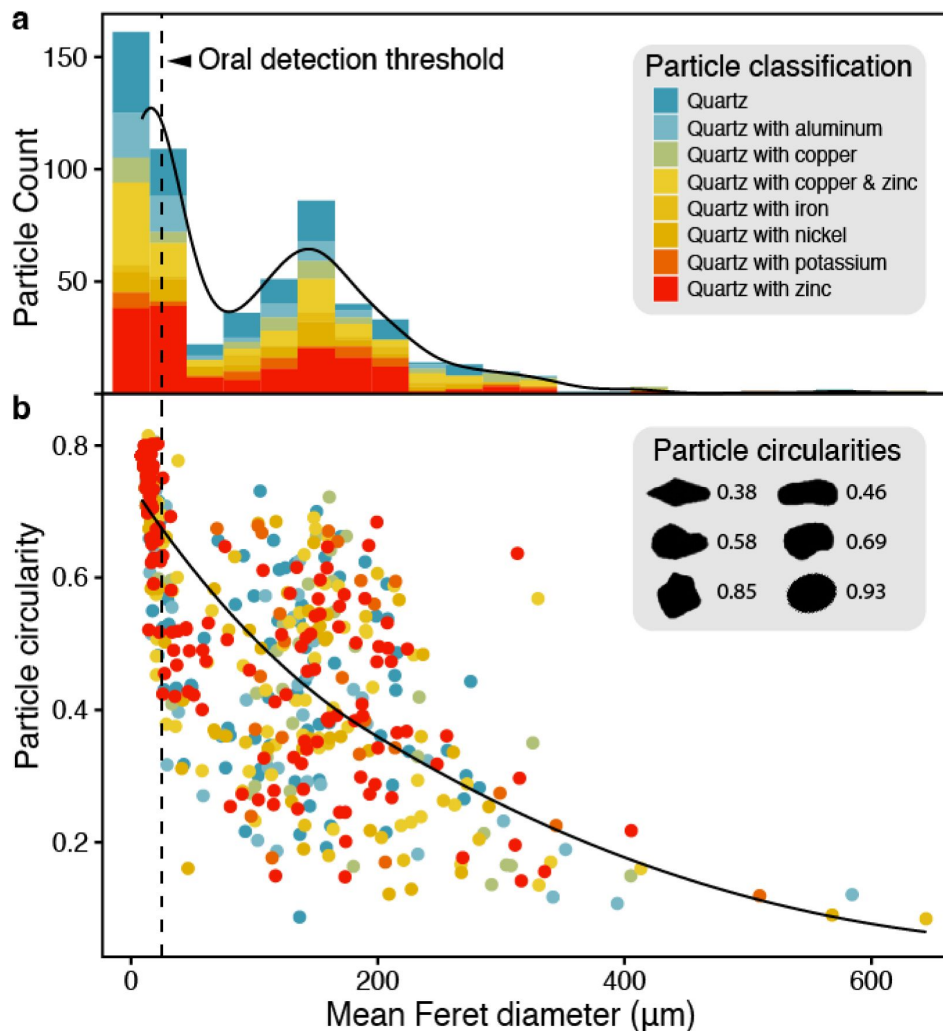


Figure 1

Variation in the mineral and physical properties of quartz particles on food surfaces.

(A) Particle sizes followed a bimodal distribution, with most particles featuring metal inclusions. Nearly half the sample is $< 25 \mu\text{m}$, the “grittiness threshold” of the human oral cavity (Imai et al., 1995). (B) Circularity is a dimensionless shape factor (range: 0 to 1) based on two-dimensional microscopy and estimates for the projected area and perimeter of a given particle. Some examples are illustrated; overall, it is a convenient but imperfect proxy for sphericity, or deviations from spherical (Grace and Ebneyamini, 2021). Here, circularity varied as a function of mean Feret diameter, suggesting that larger particles hold greater potential for damaging attack angles during particle-enamel contact.

Mitigating tooth wear

Chewing undetected quartz is expected to cause severe tooth wear, but this cost can be mitigated behaviorally if food-cleaning is proficient. To test this contention, we simulated the brushing and washing actions of monkeys with cucumber slices exposed to three concentrations of sand: low ($\bar{x} = 0.2 \pm 0.2 \text{ mg mm}^{-2}$), intermediate ($\bar{x} = 0.9 \pm 0.1 \text{ mg mm}^{-2}$), and high ($\bar{x} = 1.8 \pm 0.9 \text{ mg mm}^{-2}$). We found that brushing was less efficient than washing across treatments, eliminating $76 \pm 7\%$ vs. $93 \pm 4\%$ of sand particles, respectively (**Figure S1**). It is a modest difference, perhaps, but it is freighted with fitness consequences when extrapolated over years of life (Fannin et al., 2022). It follows that monkeys should compulsively wash sand from food whenever the opportunity avails itself, a prediction that motivated a field experiment.

Results and Discussion

Our experiment (**Figure 2A**) was designed to test two concepts at once. The first pivots around intentionality, a thorny problem that emerged from studies of raccoons (*Procyon lotor*). Celebrated food-handlers, the submersion of food objects in water is better termed ‘dousing’ for greater haptic sensation, not washing with the intention of removing surface contaminants (Lyll-Watson, 1963). If the intent of monkeys is to eliminate sand, then the time devoted to brushing or washing food should vary as a positive function of sandiness. The other concept draws on observations from Koshima Island, which alluded to rank effects on individual cleaning behaviors, a pattern that is difficult to detect without controlling access to food or distance to the ocean.

We conducted 101 feeding trials, recording 1,282 food-handling events by 42 individuals. We found that monkeys were sensitive to sand on their food, responding to each treatment—low, intermediate, and high concentrations—with greater median durations (± 1 SD) of brushing (low: 0.0 ± 0.1 s; intermediate: 1.1 ± 2.0 s; high: 3.1 ± 2.0 s; **Figure 2B**) and washing (low: 0.04 ± 0.3 s; intermediate: 0.6 ± 2.0 s; high: 3.3 ± 4.3 s; **Figure 2C**). This result is important for upholding long-held assumptions of intentional cleaning. Further, we found that dominant monkeys of both sexes showed a strong propensity for food-brushing (**Figure 2B**; **Table S3**) over food-washing (**Figure 2C**; **Table S4**). This finding reverses the pattern observed on Koshima Island (Watanabe, 1994), and it raises the possibility that food-washing is an indulgence subject to diminishing returns. To explore this premise, we developed a theoretical model where the time devoted to food-cleaning is predicted to maximise the rate of sand removal as a function of handling time.

Figure 3A illustrates the fastidious nature of our study population: monkeys allocated excess time to washing and brushing—by factors of 1.5 and 3.0, respectively—beyond that predicted by the optimization of sand removal. Our model also highlights sharply divergent responses to the sunk costs of food-handling time (**Figure 3B**). Given the efficacy of washing (**Figure S1**) and time needed to carry food to the ocean ($\bar{x} = 22 \pm 15$ s; range: 5 to 78 s), there is little incentive to over-wash food (**Figure 3B**, region I). At the same time, the lowest- and highest-ranking monkeys abstained from washing altogether, choosing instead to minimize food-handling time by overbrushing their food (**Figure 3B**, region II). This tolerance for fast-diminishing returns underscores the monkeys’ strong aversion to sand; but even so, the long-term benefits of mitigating tooth wear must be balanced against urgent energetic requirements.

Disposable soma

The disposable-soma hypothesis of senescence predicts investment in the immediate needs of survival or reproduction over tooth preservation (Carranza et al., 2004). Dominant monkeys face this predicament because rapid food intake rates are integral to sustaining dominance and

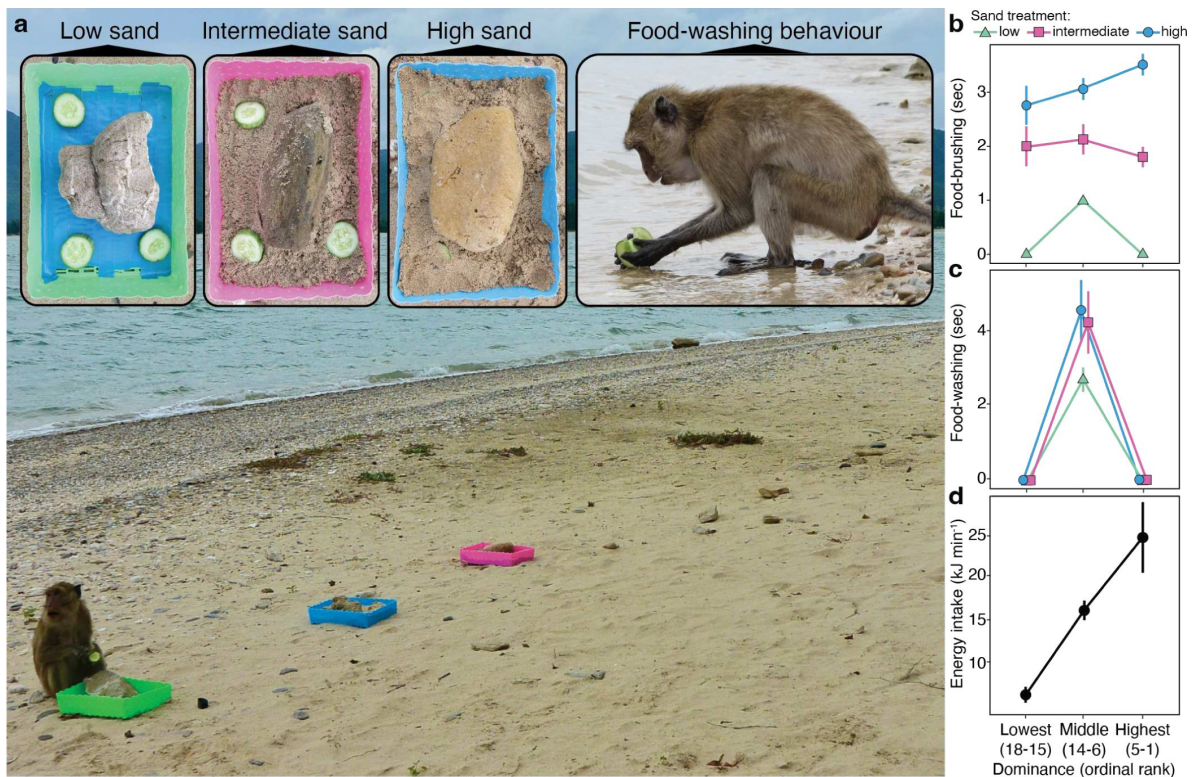


Figure 2

Experimental design and results.

(A) To elicit food-cleaning behaviors, we put sliced cucumbers in trays representing three treatments—food surfaces with low ($\bar{x} = 0.2 \pm 0.2 \text{ mg mm}^{-2}$), intermediate ($\bar{x} = 0.9 \pm 0.1 \text{ mg mm}^{-2}$), and high ($\bar{x} = 1.8 \pm 0.9 \text{ mg mm}^{-2}$) concentrations of sand—positioned 1.5 m apart and 15 m from the ocean. (B) Monkeys brushed sandier treatments for longer durations, $\chi^2 (2, n = 575 \text{ food-handling bouts}) = 185.7, p < 0.0001$ with no difference across dominance ranks (**Table S3**). (C) Monkeys washed sandier treatments for longer durations, $\chi^2 (2, n = 362 \text{ food-handling bouts}) = 50.5, p < 0.0001$, and we detected an interaction effect with dominance rank, $\chi^2 = 6.4, p = 0.04$ (**Table S4**). Raw data for panels B and C are illustrated in **Figure S2**. (D) Energy intake rates also varied as function dominance (ANOVA; $F_{2,104} = 4.3; p = 0.02$). Symbols represent mean values and whiskers $\pm 1 \text{ s.e.}$ Photos by Amanda Tan.

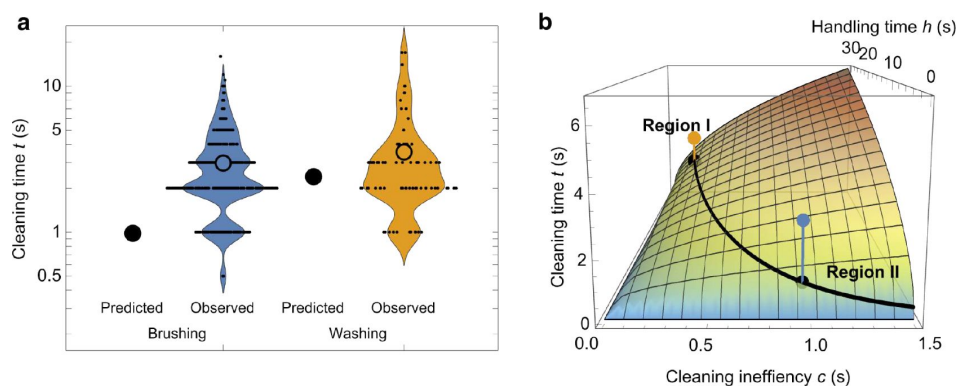


Figure 3

Predicted and observed cleaning times.

a, Predicted time (large filled points) vs. observed times (violin plots) for brushing and washing food (note log scale). b, Predicted cleaning time as a function of cleaning inefficiency c , and handling time h , with predicted values (black points) for brushing and washing based on observed cleaning inefficiencies and handling times. The coloured points (as in panel a) represent observed cleaning times. The tradeoff between longer food handling times and efficient cleaning (oceanside food-washing; Region I) and shorter handling times and inefficient cleaning (immediate food-brushing; Region II) is depicted by the black curve.

accruing reproductive success. For dominant males, energy intake determines their ability to sustain consortships during mating (Higham et al., 2011 [↗](#)); and, for dominant females, it affects practically every measure of fitness (Alberts, 2019 [↗](#); Cooper et al., 2022 [↗](#)). Our findings suggest that dominant monkeys refrained from washing to maximize short-term energy intake (**Figure 2D** [↗](#)). In short, they prioritized pressing energetic needs over the long-term benefits of tooth preservation—a ‘live fast, die young’ life-history strategy. This view of teeth as disposable soma may explain why dominant male monkeys experience faster senescence and earlier mortality (Anderson et al., 2021 [↗](#)). Estimating fitness consequences is beyond the scope of our study, but our findings suggest that a prolonged life is also subject to diminishing returns.

Paleo matter(s)

The full extent of sand-mediated tooth wear is unknown for our study population, but it is probably extreme among the highest-ranked individuals. If affirmed, the findings could affect our views of the hominin fossil record by challenging the assumption that dietary variability is the principal cause of variable dental wear. Some species, notably *Paranthropus boisei*, had ready access to water, which raises the possibility that they—like many primate species—assiduously washed their food, an essential behavior if their diet featured gritty underground plant tissues (Wrangham et al., 2009 [↗](#); Fannin et al., 2021 [↗](#)). Other species, notably *P. robustus*, have extremely variable levels of tooth pitting (Peterson et al., 2018 [↗](#)), which could reflect, at least partially, the absence of extensive wetlands (Herries et al., 2010 [↗](#)) coupled with interindividual variation in food-cleaning behaviors. Tellingly, the dental wear observed on the macaques of Koshima Island bears striking similarities to the hominin fossil record (Towle et al., 2022 [↗](#)), suggesting that populations of food-cleaning monkeys are a valuable model system that warrant further study.

Significance

Our study leverages a new method in ecological research to provide the first analysis of siliceous particles on primate foods. Our experiment probes the behavioral economics of wild monkeys, revealing a strong aversion to sandy foods. Yet, the monkeys behaved irrationally when cleaning their foods, allocating excess time than predicted by an optimization model. Some individuals fell victim to the sunk cost fallacy by over-washing their foods, whereas dominant monkeys abstained from washing altogether, seemingly sacrificing their teeth at the altar of high rank, a social status that relies rapid food intake. Our results support the disposable-soma hypothesis for senescence and kick the tires of a treasured assumption in paleoanthropology.

Methods and Materials

Study site and population

Koram Island (12.242°, 100.009°) lies ~1 km offshore in the Gulf of Thailand and within Khao Sam Roi Yot National Park, Prachuap Khiri Khan, Thailand. It has an area of 0.45 km² and a coastline of 3.5 km. The habitat—limestone karst blanketed with a dense flora of dwarf evergreen trees and deciduous scrub, and encircled by rocky shore and sandy beaches—supports a population of ca. 75 long-tailed macaques described as hybrids at the subspecies taxonomic level (*Macaca fascicularis aurea* x *M.f. fascicularis*) (Gumert et al., 2019 [↗](#)). The animals are well habituated to human observers due to regular tourism and sustained study since 2013 (Tan et al., 2018 [↗](#)).

Rank determination

Macaques form multi-male multi-female (polygynandrous) social groups with individual dominance hierarchies. Among females, this hierarchy is strictly linear and stable through time (van Noordwijk and van Schaik, 1999 [↗](#)). To determine the rank-order of adults, we recorded

dyadic agonistic interactions and their outcomes (i.e., aggression, supplants, and silent-bared-teeth displays of submission) during 5-min focal follows of individuals based on a randomized order of continuous rotation (Tan et al., 2018 [DOI](#)). In some cases, these data were supplemented with ad libitum observations. This protocol existed during five years (2013-2018) of continual observations before we conducted our experiment in July-August 2018. To determine the effects of dominance rank on individual foodcleaning propensities, and to standardize the data by sex, we followed the methods of Levy et al. (2020) [DOI](#). We calculated ordinal ranks between 1 (highest) and n (lowest), where n is the number of animals aged ≥ 5 years in each hierarchy. We analyzed male (n = 8) and female (n = 16) dominance hierarchies together in the same statistical models. We chose ordinal ranks because they best approximate competitive regimes during density-dependent competition (Levy et al., 2020 [DOI](#)). This type of feeding competition reflects those of our experimental trials and the natural conditions on Koram Island, where preferred resources are limited (Luncz et al., 2017 [DOI](#)).

Measuring sand

To quantify the amount of sand on food surfaces, whether provisioned by tourists (cucumbers, melon, pineapple) or used in our experiment (sliced cucumbers), we applied a quick-drying liquid polymer—granulated plastic (Pioloform BL 16; Wacker-Chemie GMBH, Munich, Germany) mixed with ethanol (18% plastic to 82% ethanol, by weight)—to each food item. When dried, we peeled and stored the sand-infused film for analysis. The advantages of this method are twofold: the removal of exogenous particulate matter is extremely thorough; and the plastic does not detach biogenic silica such as trichomes or phytoliths (Hinton et al., 1996 [DOI](#)). In the lab, we dissolved each peel in ethanol and separated the sand by centrifugation, producing a pellet. We dried and weighed the pellet, dividing the mass by the surface area of the food object, which we calculated from digital photographs imported into ImageJ v. 1.52. This method produces an estimate of exogenous particulate mass per area (mg mm^{-2}), allowing direct comparison of apples and oranges.

To measure the elemental and physical properties of sand, we dispersed and filtered the pellets in water using a 0.2- μm isopore membrane filter, which we submitted for scanning electron microscopy (SEM) and energy dispersive X-ray spectroscopy (EDS). To establish the parameters for multi-field and bulk analysis, we imaged a representative area of the filter at multiple magnifications and performed discrete particle analysis. 50 $\times$ magnification allowed for statistically significant representation of particle number and size range (allowing a 5 μm lower particle size range in analysis). All discrete particle analyses indicated silicon rich particles and composition distribution bins were established to include dominant accompanying elements. After establishing these parameters, we initiated multi-field automated analysis using six fields of view of the debris field (at 50 $\times$). A composition classification was assigned to each particle and data sorted by composition classification and particle size (particle sizing was binned using standard Feret maximum parameter). The sizing bins are standard ISO-16232 size classes.

Experimental design

To elicit food handling behaviors and determine individual cleaning preferences, we put three cucumber slices in each of three trays (20 $\times$ 30 $\times$ 10 cm) and manipulated the amount of contaminating sand. In the low-sand treatment, we put cucumber slices in a tray without sand; however, the presence of some aeolian sand was unavoidable. In the intermediate-sand treatment, we lined the tray with sand and put the cucumber slices on the surface. In the high-sand treatment, we buried cucumber slices completely (Figure 2A [DOI](#)). Trays were placed 1.5 m apart and 15 m from the ocean, and we randomized the color and sequence of trays across trials.

Trials began when one or more monkeys approached the trays and ended when the animals finished every cucumber slice or abandoned the experiment (range: 10 s to 14 min). We used video recordings to determine the onset and offset of individual food-handling bouts, beginning from

initial contact with a cucumber slice and ending when the final slice, in its entirety, entered the mouth. Within each bout, we determined the duration of brushing and washing behaviors, defining each from the onset of serial stereotypical forelimb movements to the moment of oral ingestion. We estimated energy intake rates by calculating the number of cucumber slices consumed during each food-handling bout, and multiplying each slice by 1.1 kcal (source: U.S. Department of Agriculture, FoodData Central, 2019) and 4.186 kJ (Hargrove, 2007 [↗](#)). We performed 101 trials over 5 weeks and recorded 1,282 food-handling bouts by 42 individual monkeys. To minimize the potential confounding effects of dominance interactions, we analyzed trials with ≤ 3 monkeys. Thus, 935 food-handling bouts were included in GLMM statistical models, which included data on individual rank, sex, and sand treatment. If a monkey consumed a cucumber slice without brushing or washing it, the zero-second duration was included in both GLMMs.

Behavioral analyses

To model experimental variance in brushing and washing behaviors as a function of experimental treatment, sex, and rank, we fit generalized linear mixed models (GLMMs) using the *glmmTMB* package in R version 4.2.3 (Brooks et al., 2017 [↗](#)). In each GLMM, we modelled *sand treatment* (a categorical variable with three levels), *sex* (a categorical variable with two levels), and *ordinal rank* (a discrete variable ranging from 1 to 18) as fixed effects. We also incorporated two additional interaction terms as fixed effects: *sand treatment* $\times$ *ordinal rank* and *sand treatment* $\times$ *sex*. To account for experimental variance among individuals and control for pseudoreplication (because the number of feeding bouts per individual varied widely; Bolker et al. (2009) [↗](#)), we included individual ID as a random intercept. The brushing and washing data sets were whole-number counts (seconds) with means < 5 . The distributions were right-skewed with high concentrations of biologically-meaningful zeros (Martin et al., 2005 [↗](#)) (i.e., instances of food-handling without any cleaning behavior). Thus, we fit four separate models in *glmmTMB* to account for these non-normal spreads: standard GLMMs with a log-link function and either a (1) Poisson or (2) negative binomial error distribution (default = `nbinom2` in *glmmTMB*); or zero-inflated generalised linear mixed models (ZIGLMM) with a logit-link function, a single zero-inflation parameter applying to all observations, and either a (3) Poisson or (4) negative binomial error distribution. We then determined the best fit model using delta AIC values in the *bbmle* package in R.

ΔAIC values indicated that, for both brushing (Table S1 [↗](#)) and washing (Table S2 [↗](#)), negative binomial GLMMs without zero-inflation were the best-fitting models. We validated each model by calculating dispersion statistics (χ^2 /degrees of freedom) (Zuur and Ieno, 2016 [↗](#)). Dispersion statistics for the brushing model ($\chi^2 = 541.9$; residual degrees of freedom = 564; χ^2 /rdf = 0.96, $p = 0.74$, one-sided test) and the washing model ($\chi^2 = 174.2$; residual degrees of freedom = 351; χ^2 /rdf = 0.50, $p = 1.00$, one-sided test) failed to detect overdispersion in either case. We report the fixed effects tests for each GLMM in Tables S3 [↗](#) and S4 [↗](#) as Analysis of Deviance Tables (Type II Wald chi square tests, one-sided) along with χ^2 values, degrees of freedom, and p-values (one-sided tests). For all statistical analyses, $\alpha = 0.05$.

Optimal cleaning time model

To model the optimization of sand removal before consumption, we accounted for two distinct temporal periods: handling time h , which includes an assessment time and pre-cleaning time, and the cleaning time t . Assessment time (set as a constant 1 second) includes visual fixation on a food object and forelimb extension before contact, whereas pre-cleaning time represents all handling activities that precede cleaning. During brushing, the pre-cleaning time was essentially nil (zero seconds), but washing required travel from the experimental treatments to the ocean, requiring longer pre-cleaning times ($\bar{x} = 22 \pm 15$ s; range: 5 to 78 s). We assumed that the proportion of sand removed from each cucumber follows the saturating relationship $g(t) = t/(c + t)$, where c is the halfsaturation constant associated with brushing or washing. As c increases, so does the

inefficiency of a given cleaning behaviour. Given our observations that brushing removes 75% of grit in 2.97 s, and washing removes 93% of grit in 3.53 s (**Figure S1** [↗](#)), we obtain the constants $c_{brushing} = 0.99$ s and $c_{washing} = 0.26$ s, such that washing (without considering handling costs) is the most efficient strategy. The rate of grit removal is then given by $R(t) = g(t)/(h + t)$, which reaches a maximum at the optimal cleaning time $t^* = \sqrt{ch}$. For brushing and washing cleaning strategies, we obtain the expected optimal cleaning times $t^*_{brushing} = 0.98$ s, and $t^*_{washing} = 2.39$ s (**Figure 3a** [↗](#)), respectively. These optimal cleaning times are defined exclusively with respect to maximising the rate of grit removal, without considering the potentially cascading effects of these strategies on fitness.

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Additional Declarations:

The authors declare no competing interests.

Supplemental Materials

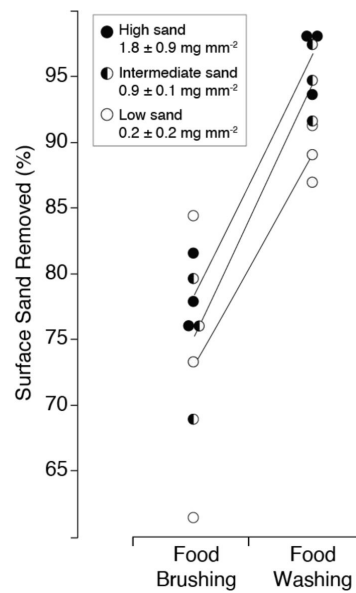


Fig S1.

Comparison of cleaning effectiveness.

We simulated the brushing and washing behaviours of our study animals using the same three treatments of cucumber slices in our experiment. Brushing was less efficient than washing across all treatments, eliminating an average $76 \pm 7\%$ vs. $93 \pm 4\%$ of surface sands, respectively.

Fig S2.

Individual food-cleaning events.

(A) Monkeys put more time into brushing sandier treatments, χ^2 (2, n = 575 food-brushing events) = 185.7, p < 0.0001) with no difference across dominance ranks (Table S3). (B) Monkeys put more time into washing sandier treatments, χ^2 (2, n = 362 food-washing events) = 50.5, p < 0.0001), but we detected an interaction effect with dominance rank, χ^2 = 6.4, p = 0.04 (Table S4).

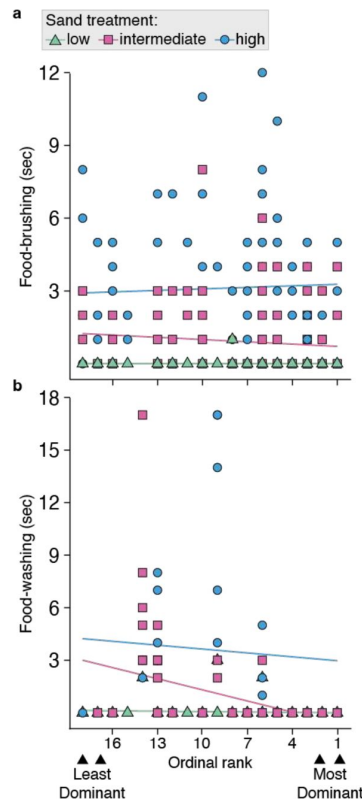


Table S1

delta AIC model results for brushing models - testing different model distributions and potential zero-inflation for brushing model (n = 575 observations)

Model	dAICc	Degrees of Freedom
Fit: Negative Binomial	0.0	11
Fit: Poisson	0.8	10
Fit: Zero-Inflated Poisson	1.1	11
Fit: Zero-Inflated Negative Binomial	1.7	12

Model	dAICc	Degrees of Freedom
Fit: Negative Binomial	0.0	11
Fit: Zero-Inflated Negative Binomial	2.1	12
Fit: Poisson	20.2	10
Fit: Zero-Inflated Poisson	21.0	11

Table S2

delta AIC model results for washing models - testing different model distributions and potential zero inflation for washing models (n = 362 observations)

Table S3

Fixed effects for food-brushing (n = 575 events by animals with known rank), analysis of deviance (Type II Wald Chi Square Tests)

Fixed Effect	χ^2	Degrees of Freedom	p (one sided)
Grit treatment	185.74	2	p< 0.0001
Ordinal rank * grit treatment	2.57	2	0.2765
Ordinal rank	0.01	1	0.3253
Sex	0.09	1	0.7703
Sex * grit treatment	0.13	2	0.9368

Table S4

Fixed effects for food-washing (n = 362 events by animals with known rank), analysis of deviance (Type II Wald Chi Square Tests)

Fixed Effect	χ^2	Degrees of Freedom	p (one sided)
Grit treatment	50.45	2	p< 0.0001
Ordinal rank * grit treatment	6.38	2	0.0411
Sex * grit treatment	0.77	2	0.7762
Ordinal rank	0.06	1	0.8082
Sex	0.00	1	0.9760

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Reviewer #1 (Public Review):

Summary:

In this paper, the authors had 2 aims:

(1) Measure macaques' aversion to sand and see if its' removal is intentional, as it is likely in an unpleasurable sensation that causes tooth damage.

(2) Show that or see if monkeys engage in suboptimal behavior by cleaning foods beyond the point of diminishing returns, and see if this was related to individual traits such as sex and rank, and behavioral technique.

They attempted to achieve these aims through a combination of geochemical analysis of sand, field experiments, and comparing predictions to an analytical model.

The authors' conclusions were that they verified a long-standing assumption that monkeys have an aversion to sand as it contains many potentially damaging fine-grained silicates and that removing it via brushing or washing is intentional.

They also concluded that monkeys will clean food for longer than is necessary, i.e. beyond the point of diminishing returns, and that this is rank-dependent.

High and low-ranking monkeys tended not to wash their food, but instead over-brushed it, potentially to minimize handling time and maximize caloric intake, despite the long-term cumulative costs of sand.

This was interpreted through the *disposable soma hypothesis*, where dominants maximize immediate needs to maintain rank and increase reproductive success at the potential expense of long-term health and survival.

Strengths:

The field experiment seemed well-designed, and their quantification of physical and mineral properties of quartz particles (relative to human detection thresholds) seemed good relative to their feret diameter and particle circularity (to a reviewer who is not an expert in sand). The *Rank Determination* and *Measuring Sand* sections were clear.

In achieving Aim 1, the authors validated a commonly interpreted, but unmeasured function, of macaque and primate behavior-- a key study/finding in primate food processing and cultural transmission research.

I commend their approach in developing a quantitative model to generate predictions to compare to empirical data for their second aim.

This is something others should strive for.

I really appreciated the historical context of this paper in the introduction, and found it very enjoyable and easy to read.

I do think that interpreting these results in the context of the *disposable soma hypothesis* and the potential implications in the *paleolithic matters* section about interpreting dental wear in the fossil record are worthwhile.

Weaknesses:

Most of the weaknesses in this paper lie in statistical methods, visualization, and a missing connection to the marginal value theorem and optimal foraging theory.

I think all of these weaknesses are solvable.

The data and code were not submitted. Therefore I was unable to better understand the simulation or to provide useful feedback on the stats, the connection between the two, and its relevance to the broader community.

(1) Statistics:

(a) AIC and outcome distributions

The use of AIC for hierarchical models, and models with different outcome distributions brought up several concerns.

The authors appear to use AIC to help inform which model to use for their primary analyses in Tables S1 and S2. It is unclear which of these models are analyzed in Tables S3 and S4.

AIC should not be used on hierarchical models, and something like WAIC (or DIC which has other caveats) would be more appropriate.

Also, using information criteria on Mixture Models like Negative Binomials (aka Gamma-Poisson) should be done with extreme caution, or not at all, as the values are highly sensitive to the data structure.

Some researchers also say that information criteria should not be used to compare models with different outcome distributions - although this might be slightly less of a concern as all of your models are essentially variations on a Poisson GLM.

Discussion on this can be found in McElreath Statistical Rethinking (Section 12.1.3) and Gelman et al. BDA3 (Chapter 7).

Choosing an outcome distribution, based on your understanding of the data generating process is a better approach than relying on AIC, especially in this context where it can be misleading.

(b) Zeros

I also had some concerns about how zeros were treated in the models.

In lines 217-218, they mentioned that "if a monkey consumed a cucumber slice without brushing or washing it, the zero-second duration was included in both GLMMs."

This zero implies no processing and should not be treated as a length 0 duration of processing.

This suggests to me that a zero-inflated poisson or zero-inflated negative binomial, would be the best choice for modelling the data as it is essentially a 2-step process:

- (i) Do they process the cucumber at all?
- (ii) If so do they wash or brush, and how is this predicted by rank and treatment?

(2) Absence of Links to Foraging Theory

Optimal cleaning time model: the optimality model was not well described including how it was programmed. Better description and documentation of this model, along with code (Mathematica judging from the plot?) is needed.

There seems to be much conceptual and theoretical overlap with foraging theory models that were not well described - namely the *marginal value theorem (Charnov (1976), Krebs et al. (1974),) and its subsequent advances* (see <https://doi.org/10.1016/j.jaa.2016.03.002> and <https://doi.org/10.1086/283929> for examples).

In the suggestions, I attached the R code where I replicated their model to show that it is *mathematically identical to the marginal value theorem*. This was not mentioned at all in the text or citations.

This is a well-studied literature since the 1970's and there is a history of studies that compare behavior to an optimality model and fail (or do find) instances where animals conform or diverge with its predictions (<https://doi.org/10.1146/annurev.es.15.110184.002515>). This link should be highlighted, and interpreting it in that theoretical context will make it more broadly applicable to behavioral ecologists.

The data was subsetting to include instances where there were < 3 monkeys present to avoid confounds of rank, but it is important to know that optimal behavior might vary by individual, and can change in a social context depending on rank (see <https://doi.org/10.1016/j.tree.2022.06.010>). Discussion of this, and further exploration of it in the data would strengthen the overall contribution of this manuscript to the field, but I understand that the researchers wish to avoid that in this paper for it is a complex topic, which this dataset is uniquely suited to address.

(3) Interpretation and validity of model relative to data

In lines 92-102, they present summary statistics (I think) showing that time spent brushing and washing is consistent with washing or brushing to remove sand.

In the **mitigating tooth wear** section (line 73) and corresponding Figure S1 showing surface sand removed, more detail about how these numbers were acquired, and statistical modelling, is needed.

This is important as uncertainty and measurement error around these metrics are key to the central finding and interpretation of Aim 2 in this paper.

It appears that the researchers simulated the monkey's brushing and washing behaviors (similar to <https://doi.org/10.1007/s10071-009-0230-3>).

How many researchers simulated monkey behavior and how many times?

What are the repeat points in Figure S1?

What is the number of trials or number of people?

This effect appears stronger for washing than brushing as well - if so, why?

More info about this data, and the uncertainty in this is important, as it is key to the second central claim of this paper.

The estimates of removing between 76% +/- 7 and 93% +/- 4 of sand (visualized in Figure S1), are statistical estimates.

I would find the argument more convincing if after propagating for the uncertainty in handling in sand removal rates, and the corresponding half-saturation constants, if this processing for food is too long, after accounting for diminishing returns held true. It is very possible that after accounting for uncertainty and variation in handling time and removal rates, the second result may not hold true.

I was not able to convince myself of this via reanalysis as the description of the data in the text was not enough to simulate it myself.

Essentially, this would imply that in Figure 3 the predicted value would have some variation around it (informed by boundary conditions of time being positive, and percents having floors and ceilings) and that a range of predicting cleaning times (optimal give-up times) would be plotted in Figure 3.

This could be accomplished in a Bayesian approach, Or by simply plotting multiple predictions given some confidence interval around, c and h.

<https://doi.org/10.7554/eLife.98520.1.sa2>

Reviewer #2 (Public Review):

Summary:

This field experiment aimed to assess what motivates macaque monkeys to clean food items prior to consumption and the relative costs and benefits of different cleaning approaches (manually brushing sand from food versus dousing food items in water). The experiment teases apart if/how the benefits of these approaches are mediated by the amount of debris on food and the monkeys' rank in terms of the costs of consuming sand versus the time and energy required to remove it. The authors not only examined the behavioral responses of wild macaques to three conditions of food sand contamination but also tested the relative costs of consuming different levels and sizes of sand particulates. Through this, the authors propose considerations of the macaques' motivations to clean food and the balance they take in energetic gains from consuming food versus the costs of cleaning food and consuming sand. Their data reveal that food washing is more effective in removing sand, but more costly than manually brushing off sand. This study also revealed that only mid-ranked monkeys washed their food, while high and low-ranked monkeys were more likely to remove sand via brushing it off food with their hands.

Strengths:

This study provides a very in-depth consideration of the motivations of macaques to clean their food, and the relative costs and benefits of different food cleaning techniques. Not only did the study test the behavior of wild macaques via a simple yet elegant field study, but they also performed a detailed analysis of the sand particulates to understand the level of potential tooth wear that consuming it could result in. By relying on a wild group of macaques that have been part of a long-term study site, the team also had detailed behavioral data on the population to allow for rank assessments of the animals. This comprehensive study provides important foundational information for a better understanding of how and why macaques clean food, that inform existing and future considerations of this as a potential cultural behavior.

Weaknesses:

As currently written, the paper does not provide sufficient background on this population of animals and their prior demonstrations of food-cleaning behavior or other object-handling behaviors (e.g., stone handling). Moreover, the authors' conclusions focus on the behavior of high-ranked animals, but subordinate animals also showed similar behavioral patterns and they should be considered in more detail too.

<https://doi.org/10.7554/eLife.98520.1.sa1>

Reviewer #3 (Public Review):

This paper provides evidence that food washing and brushing in wild long-tailed macaques are deliberate behaviors to remove sand that can damage tooth enamel. The demonstration of the immediate functional importance of these behaviors is nicely done. However, the paper also makes the claim that macaques systematically differ in their investment in food cleaning because of rank-dependent differences in their costs and benefits. This latter conclusion is not, in my view, well-supported, for several reasons.

First, as is typical in many primate studies, the authors construct sex-specific ordinal rank hierarchies. This makes sense since hierarchies for males and hierarchies for females are determined by different processes and have different consequences. However, if I understand it correctly, they are then lumped together in all statistical analyses of rank, which makes the apparent rank effect very difficult to understand. The challenge of interpretation is increased because there are twice as many adult females in the group as adult males, so the rank is confounded by sex (because all low-rank values are adult females).

Second, because only one social group is being studied, the conclusions about rank may be heavily driven by individual identity, not rank per se. An analysis involving replicate social groups (which granted, may be impossible here) or longitudinal data showing a change in behavior following a change in rank would be much more compelling.

Third, there is no evidence presented on the actual fitness-related costs of tooth wear or the benefits of slightly faster food consumption. Support for these arguments is provided based on other papers, some of which come from highly resource-limited populations (and different species). But this is a population that is supplemented by tourists with melons, cucumbers, and pineapples! In the absence of more direct data on fitness costs and benefits, the paper makes overly strong claims about the ability to explain its observations based on "immediate energetic requirements" (abstract), "difference...freighted with fitness consequences" (line 80), and "pressing energetic needs"/"live fast, die young" (lines 121-122--there is no evidence that tooth wear is associated with morbidity or mortality here). The idea that high-ranking animals are "sacrificing their teeth at the altar of high rank" seems extreme.

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